

GORILLA (Male vs Female)

Low Evolvability

evolutionary potential arrow

Trapped in Local Optima

High Integration / Morphological Constraints

High Evolvability

Evolvability enables exploration of distant morphospace

Phenotypic variation is the prerequisite

Modular Design / Reduced Constraints

Access to Novel Morphologies

A detailed illustration of a bat skull, viewed from the side. The braincase is colored purple, and the jaw, showing a series of sharp teeth, is colored green. The illustration uses fine lines to depict the texture and structure of the bone.

$$\text{conditional evolvability}_{(\text{SD})} = \text{evolvability}_{(\text{SD})} \cdot \text{integration}$$

Direction of Sexual Dimorphism

- Greater sexual dimorphism is associated with higher evolvability

A scatter plot showing the relationship between Sexual Dimorphism (norm) on the x-axis and Conditional Evolvability on the y-axis. The x-axis ranges from 0.0 to 1.5, and the y-axis ranges from 0.00 to 0.05. Two data series are plotted: 'Mean' (red circles) and 'Sexual Dimorphism' (teal triangles). The 'Mean' series shows low values, mostly below 0.01. The 'Sexual Dimorphism' series shows higher values, with several points between 0.01 and 0.05, and a few outliers up to 0.05. A legend in the top left corner identifies the series.

A scatter plot showing the relationship between Morphological Integration (x-axis) and Evolvability (sd) (y-axis). The x-axis ranges from 0.85 to 1.00, and the y-axis ranges from 0.00 to 0.20. A blue regression line indicates a positive correlation, with $R^2 = 0.03$. A grey shaded area represents the confidence interval around the regression line. Data points are represented by black dots.

Figure 2 consists of three scatter plots arranged horizontally, each showing the relationship between 'Sexual Dimorphism (norm)' on the x-axis and 'Evolvability (SD)' on the y-axis. The plots are labeled with values 0.1, 0.5, and 0.9, likely representing different trait categories or thresholds. Each plot contains numerous black data points, a solid blue regression line, and a light gray shaded area representing the confidence interval. The first plot (0.1) shows a strong positive correlation with $R^2 = 0.54$ and $P < 0.001$. The second plot (0.5) shows a moderate positive correlation with $R^2 = 0.32$ and $P < 0.001$. The third plot (0.9) shows a weak positive correlation with $R^2 = 0.08$ and $P = 0.013$.

Scatter plot showing the relationship between Sexual Dimorphism (norm) on the x-axis and Evolvability (sd) outside $P_{\max}$ on the y-axis. The data points are black dots. A blue regression line is shown, along with a grey shaded confidence interval. The coefficient of determination is $R^2 = 0.36$.

- Sexual dimorphism tends to evolve along highly evolvable directions
- Integration alone does not explain variation in evolvability among species
- Measurement error does not substantially alter the observed patterns
- Sexual dimorphism is frequently aligned with **P_{max}**
- Identify additional predictors of evolvability in the direction of sexual dimorphism.
- Test whether these patterns are phylogenetically widespread
- Examine the consequences of removing **P_{max}** on evolvability estimates.